

A Bayesian Critique of a Rank-Based Method for Surrogate Marker Evaluation

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SUMMARY: In clinical trials, surrogate markers are used to replace primary outcomes that are difficult, costly, or time-consuming to collect. When valid, they can accelerate treatment evaluation by reliably capturing the causal effect of treatment on clinical benefit. Parast, Cai, and Tian (2024, *Biometrics*) recently proposed a rank-based method for surrogate marker evaluation in small-sample settings, exploiting its nonparametric nature and minimal modeling assumptions. However, this method has notable limitations: its target estimand lacks a clear causal interpretation, it exhibits low statistical power, and it is not sufficiently robust across different data-generating processes (DGPs). We propose BSET (Bayesian Surrogate Evaluation Test), a Bayesian imputation-based approach that addresses these shortcomings by targeting a causally interpretable estimand and improving power through covariate adjustment. We derive analytical expressions for both estimands under Gaussian DGPs, showing they can differ substantially even in simple settings, and introduce a principled criterion for selecting the surrogate validation threshold based on a Bayes factor argument. We evaluate BSET through a simulation study and illustrate the method using data from the Diabetes Control and Complications Trial.

KEY WORDS: Bayesian Inference; Causal Inference; Clinical Trials; Mann-Whitney Statistic; Surrogate Markers.

1. Introduction

Surrogate markers are often employed in clinical trials to replace primary outcomes that may be difficult, costly, or time-consuming to measure (Elliott, 2023a). These markers can accelerate the evaluation of promising new treatments, provided that they reliably capture the causal relationship between treatment and true clinical benefit. Notably, in the United States, the Food and Drug Administration’s (FDA) Accelerated Approval Program, established in 1992, allows new treatments to receive early approval based on demonstrated effects on surrogate markers, and many drugs have been approved, and continue to be approved, through this pathway (U.S. Food and Drug Administration, 1992, 2025).

A valid surrogate can substantially reduce trial burden, duration, and cost while still providing meaningful information about treatment efficacy, thereby enabling more timely access to effective treatments. However, establishing *surrogate validity* is a challenging statistical problem, requiring methods that are both robust and interpretable. On this account, surrogate evaluation has been an active area of research, particularly because the FDA does not provide a formal statistical definition of a valid surrogate. A recent review by Elliott (2023b) summarizes the state of proposed and currently used methods, including the proportion of treatment effect explained (PTE) framework (Freedman et al., 1992; Wang and Taylor, 2002; Parast et al., 2016), the principal stratification framework (Frangakis and Rubin, 2002; Gilbert and Hudgens, 2008), and the meta-analytic framework (Buyse and Molenberghs, 1998; Burzykowski et al., 2005; Huang and Gilbert, 2011).

Unfortunately, these methods often perform poorly in small trials, especially when using robust techniques such as kernel-based estimation within the PTE framework. To address this, Parast et al. (2024) proposed a rank-based approach for evaluating surrogate markers in such settings. Despite its attractive nonparametric nature and minimal assumptions,

this method has notable limitations, including unclear causal interpretation for its target estimand, low power, and limited robustness across DGPs.

In this paper, we propose BSET (Bayesian Surrogate Evaluation Test), an imputation-based Bayesian alternative that extends the rank-based framework of Parast et al. (2024). Our approach is grounded in Bayesian causal inference, which treats unobserved potential outcomes as latent variables and samples them from predictive distributions (Li et al., 2023). By using this perspective, our method not only provides a causal interpretation but also allows for covariate adjustment, improved efficiency, and a flexible framework for inference.

The rest of this work is organized as follows. In Section 2, we review the existing rank-based approach for surrogate evaluation, outline its limitations, and highlight how the choice of treatment effect estimands can affect conclusions about surrogate validity. In Section 3, we present our Bayesian methodology, including two parametric models and a new criterion for the surrogate validation threshold. In Section 4, we present a simulation study comparing our proposed method with the existing approach, and in Section 5, we apply our method to a clinical trial dataset. Finally, Section 6 discusses potential future extensions.

2. Methods

2.1 Notation

Let Y denote the primary outcome and S the surrogate marker. Let $Z \in \{0, 1\}$ be the treatment indicator, assigned at random, where $Z = 1$ indicates treatment and $Z = 0$ control. Without loss of generality, assume higher values of Y and S correspond to greater clinical benefit. We adopt the potential outcomes framework to define causal effects (Rubin, 1974). For each unit $i \in \{1, \dots, n\}$, let Y_{gi} and S_{gi} denote the potential outcomes for the primary and surrogate under treatment $g \in \{0, 1\}$. The potential outcomes for unit i are

$$P_i = (Y_{0i}, Y_{1i}, S_{0i}, S_{1i})^\top.$$

We assume that for each unit i , the vector of potential outcomes P_i is independently drawn from a joint distribution F_i , which may differ across units, and that the treatment assignment Z_i is independently drawn from a Bernoulli distribution, with $Z_i \perp\!\!\!\perp P_i$ by design. Only the potential outcomes corresponding to the observed treatment assignment are observed. That is, each unit i contributes $P_{1i} = (Y_{1i}, S_{1i})^\top$ when $Z_i = 1$, or $P_{0i} = (Y_{0i}, S_{0i})^\top$ when $Z_i = 0$. The hierarchical process that generates the data is summarized as follows:

$$\begin{aligned} (Y_i, S_i)^\top &= \begin{cases} P_{1i}, & \text{if } Z_i = 1, \\ P_{0i}, & \text{if } Z_i = 0, \end{cases} \quad i = 1, \dots, n, \\ P_i &\overset{\text{ind.}}{\sim} F_i, \quad i = 1, \dots, n, \\ Z_i &\overset{\text{i.i.d.}}{\sim} \text{Bernoulli}(p), \quad i = 1, \dots, n, \end{aligned} \tag{1}$$

where p is the known treatment assignment probability (e.g., $p = 0.5$ for a balanced trial). Finally, we denote the vector of observed data for unit i as $D_i = (Y_i, S_i, Z_i)^\top$ or $D_i = (Y_i, S_i, Z_i, X_i)^\top$, when covariates X_i are available.

In the next section, we describe the existing rank-based approach for evaluating a surrogate marker and call attention to its key limitations.

2.2 Existing Frequentist Rank-Based Approach

The main idea of Parast et al. (2024) is that if S is a valid surrogate for Y , then the treatment Z should have comparable effects on both, on a probabilistic scale that is invariant to differences in the measurement units of Y and S . Their method focuses on small-sample settings, where nonparametric kernel methods are infeasible and parametric models are difficult to validate. Motivated by this, they used the following probabilistic index

$$U_Y = \mathbb{P}(Y_{1i} > Y_{0j}) + \frac{1}{2}\mathbb{P}(Y_{1i} = Y_{0j}), \tag{2}$$

as a measure of the treatment effect on the primary outcome Y , where i and j index two different units. This quantity is often referred to as the Mann-Whitney parameter (Mann

and Whitney, 1947; Thas et al., 2012; Hoeffding, 1992; Buyse, 2010), as it is the estimand targeted by the corresponding hypothesis test. In this work, we restrict our attention to continuous outcomes, thereby excluding ties in Equation (2). Therefore, a positive treatment effect on Y is concluded if $U_Y > 0.5$. The same reasoning applies to the surrogate S , with U_S defined analogously. The discrepancy in treatment effects is then measured by their difference $\delta = U_Y - U_S$. Based on this definition, Parast et al. (2024) propose that, if S is a valid surrogate for Y , δ should be small, leading to the following hypothesis test

$$\begin{cases} H_0 : \delta \geq \varepsilon, \\ H_1 : \delta < \varepsilon, \end{cases}$$

where ε is a pre-specified threshold for surrogate validation.

To estimate the parameter of interest δ , Parast et al. (2024) compute $\hat{U}_Y$ and $\hat{U}_S$ using plug-in estimators, commonly known as the Mann–Whitney U-statistics:

$$\begin{aligned} \hat{U}_Y &= \frac{1}{n_1 n_0} \sum_{i=1}^{n_1} \sum_{j=1}^{n_0} \mathbb{1}(Y_i > Y_j), \\ \hat{U}_S &= \frac{1}{n_1 n_0} \sum_{i=1}^{n_1} \sum_{j=1}^{n_0} \mathbb{1}(S_i > S_j), \end{aligned}$$

where n_1 and n_0 are the treated and control sample sizes, and then estimate δ by $\hat{\delta} = \hat{U}_Y - \hat{U}_S$. A confidence interval for δ follows from the central limit theorem for U-statistics and closed-form variance estimators for linear combinations of U-statistics. Parast et al. (2024) conclude that S is valid if the upper bound of this interval is less than the pre-specified threshold ε .

2.3 Limitations of the Existing Approach

The nonparametric approach of Parast et al. (2024) suffers from several notable limitations. Firstly, it is worth questioning whether the treatment effect quantities considered truly capture the causal targets. In the causal inference literature, rank-based statistics have in fact been criticized for lacking a clear causal interpretation (Imbens and Rubin, 2015). For example, the Mann–Whitney test assesses whether two variables are equal in distribution, but

this condition is not necessarily informative about causal effects. To illustrate this point, note that since U_Y represents the probability that a randomly selected treated unit has a higher Y than an independently selected control, it captures global distributional differences between groups, rather than a directly interpretable causal effect. This motivates the consideration of alternative estimands that more directly target causal effects, such as

$$V_Y = \mathbb{P}(Y_{1i} > Y_{0i}),$$

which represents the probability that a randomly selected unit has a better outcome Y under treatment than under control. If we define V_S analogously for the surrogate S , then we can measure the discrepancy in treatment effects via $\theta = V_Y - V_S$. Notably, U_Y and U_S are not equivalent to V_Y and V_S , as discussed in detail by Fay et al. (2018). This difference propagates directly to the discrepancy parameters δ and θ . As we illustrate in Section 2.4, they can differ substantially even under relatively simple and well-behaved DGPs, and therefore the choice between them may lead to opposing conclusions about surrogate validity.

Secondly, the assumptions underlying the Mann–Whitney statistic may be too restrictive. In particular, the test assumes potential outcomes for each observation are independent and identically distributed. The identical distribution assumption is often unrealistic, as potential outcomes are typically influenced by covariates that vary across units.

Moreover, the numerical studies in Parast et al. (2024) show that their procedure achieves high coverage across different settings, but suffers from low power in testing surrogate validity. Approaches that prioritize higher power, even if they rely on parametric assumptions, may better detect valid surrogates, particularly in small-sample settings. Along these lines, leveraging baseline covariates is a natural strategy to increase power, as their inclusion improves efficiency and power by reducing residual outcome variability unrelated to the treatment (Zhang et al., 2008; Colantuoni and Rosenblum, 2015; Zhang and Ma, 2019). However, the rank-based framework of Parast et al. (2024) does not readily accommodate

covariate adjustment, and incorporating baseline information would likely require a rank regression-type approach, representing a substantial departure from the original framework.

Hence, in Section 3, we describe our Bayesian imputation-based methodology addressing these limitations. In particular, our approach focuses on the causal treatment effect quantities V_Y and V_S , improves power, and allows incorporation of baseline covariates. The following section clarifies the discrepancy parameters and may be skipped without loss of continuity.

2.4 Comparison between Discrepancy Parameters

As noted in Section 2.3, the parameters U_Y and V_Y are not equivalent, which carries over to δ and θ . This means that, under the same DGP, the two discrepancy parameters can take substantially different values, potentially leading to opposing conclusions about surrogate validity. To illustrate this point, it is useful to consider the analytical forms of δ and θ in a simple setting where the potential outcomes are drawn from a Gaussian distribution whose parameters depend on a baseline covariate X :

$$\begin{aligned} P_i &\stackrel{\text{ind.}}{\sim} \mathcal{N}_4(\mu^{X_i}, \Sigma^{X_i}), \quad i = 1, \dots, n, \\ X_i &\stackrel{\text{i.i.d.}}{\sim} f, \quad i = 1, \dots, n, \end{aligned} \tag{3}$$

where μ^{X_i} and Σ^{X_i} are the mean vector and covariance matrix that depend on X_i , and f is the distribution of the covariate over some space $\mathcal{X}$. Under the model in Equation (3), θ and δ have a convenient analytical expression as expectations over the covariate distribution.

Specifically, consider the following two assumptions:

- **Conditional Orthogonality:** $\Sigma_{Y_1 Y_0}^x = \Sigma_{S_1 S_0}^x = 0, \forall x \in \mathcal{X}$.
- **Conditional Homoscedasticity:** $\Sigma_{Y_1 Y_1}^x = \Sigma_{S_1 S_1}^x = \Sigma_{Y_0 Y_0}^x = \Sigma_{S_0 S_0}^x = \frac{1}{2}, \forall x \in \mathcal{X}$.

Then, it follows that

$$\delta = \mathbb{E}_{(X_i, X_j) \sim f \times f} [g(X_i, X_j)], \quad \theta = \mathbb{E}_{X_i \sim f} [g(X_i, X_i)], \text{ where}$$

$$g(x_i, x_j) = \Phi(\mu_{Y_1}^{x_i} - \mu_{Y_0}^{x_j}) - \Phi(\mu_{S_1}^{x_i} - \mu_{S_0}^{x_j}), \quad \forall x_i, x_j \in \mathcal{X}.$$

Relaxing these assumptions, the formula for g still involves a difference of Gaussian CDFs with arguments depending on conditional variances and covariances, leading to a more cumbersome expression. Full derivations for both cases appear in Appendix Web Appendix A.

Importantly, we introduce the stronger assumptions stated above only for ease of exposition, as they allow us to highlight more clearly that θ and δ may differ substantially whenever $\mathbb{E}_{X_j \sim f} [g(X_i, X_j)]$ and $g(X_i, X_i)$ differ in expectation. To illustrate this point further, we consider two analytically tractable settings.

2.4.1 Binary Covariate Setting. Let us start by considering the simple setting where $f = \text{Bernoulli}(\frac{1}{2})$. Although this choice may seem trivial, it can still have a meaningful interpretation in a clinical trial setting (e.g. X represents smoker vs non-smoker status). To simplify the analysis even further, let us also assume treatment effect homogeneity across X :

$$\mu_{Y_1}^x - \mu_{Y_0}^x = \mu_{S_1}^x - \mu_{S_0}^x = \Delta, \quad x = 0, 1,$$

for some $\Delta > 0$. As we show in Web Appendix B, the discrepancy between θ and δ is driven by Δ and the difference between means of Y and S for control units across X groups:

$$|\theta - \delta| = \frac{1}{4} |\Phi(\Delta + d_Y) - \Phi(\Delta + d_S) + \Phi(\Delta - d_Y) - \Phi(\Delta - d_S)|,$$

where $d_Y = \mu_{Y_0}^0 - \mu_{Y_0}^1$ and $d_S = \mu_{S_0}^0 - \mu_{S_0}^1$. Figure 1a illustrates how the distance between θ and δ changes as a function of d_Y and d_S for a fixed Δ . It appears that $|\theta - \delta|$ is large if

$$|d_Y| < \Delta \text{ and } |d_S| > \Delta \quad \text{or} \quad |d_Y| > \Delta \text{ and } |d_S| < \Delta.$$

[Figure 1 about here.]

This intuition is formalized analytically in Web Appendix B, where we also show that

$$\sup_{d_Y, d_S} \{|\theta - \delta|\} = \frac{1}{2} \Phi(\Delta) - \frac{1}{4} \Rightarrow \sup_{d_Y, d_S, \Delta} \{|\theta - \delta|\} = \frac{1}{4},$$

where the first supremum is proved analytically in Web Appendix B, and the second follows from the principle of iterated suprema (Rudin, 1976). In particular, $\theta = 0$ while $|\delta|$ can be

as large as $\frac{1}{4}$, so even though S is a perfect surrogate for Y , δ may be substantially different from 0, while θ correctly reflects surrogate validity.

2.4.2 Gaussian Covariate Setting. Let us now consider a setting where $f = \mathcal{N}(m, v^2)$ and the mean potential outcomes are linear functions of the covariate:

$$\mu^x = x\boldsymbol{\beta}, \quad \forall x \in \mathbb{R}, \quad \text{where} \quad \boldsymbol{\beta} = (\beta_{Y_1}, \beta_{S_1}, \beta_{Y_0}, \beta_{S_0})^\top.$$

As we show in Web Appendix C, the general formulas for θ and δ are non-trivial functions of the slope parameter $\boldsymbol{\beta}$ and the covariate distribution parameters m and v . Then, to simplify the analysis, let us assume perfect surrogacy by imposing the following assumption:

$$\beta_{Y_1} - \beta_{Y_0} = \beta_{S_1} - \beta_{S_0} = \Delta. \quad (4)$$

This implies that the treatment effect is the same for both Y and S across all values of the covariate. Under this assumption, $\theta = 0$ and therefore $|\theta - \delta| = |\delta|$, which is given by

$$|\theta - \delta| = \left| \Phi \left(\frac{m\Delta}{\sqrt{1 + v^2 [(\beta_{Y_0} + \Delta)^2 + \beta_{Y_0}^2]}} \right) - \Phi \left(\frac{m\Delta}{\sqrt{1 + v^2 [(\beta_{S_0} + \Delta)^2 + \beta_{S_0}^2]}} \right) \right|.$$

Figure 1b illustrates how the distance between θ and δ changes as a function of β_{Y_0} and β_{S_0} , for fixed Δ , m , and v . As derived in Web Appendix C, for $m, \Delta \neq 0$ we have that

$$\sup_{\boldsymbol{\beta}, \Delta} \{|\theta - \delta|\} = \left| \Phi \left(\frac{m\sqrt{2}}{v} \right) - \frac{1}{2} \right| \Rightarrow \sup_{\boldsymbol{\beta}, \Delta, m, v} \{|\theta - \delta|\} = \frac{1}{2},$$

An analytic proof for the first supremum is given in Web Appendix C, while the second supremum follows by the principle of iterated suprema (Rudin, 1976), and it is approached when $|\frac{m}{v}| \gg 0$, which occurs when the Gaussian covariate is concentrated away from zero.

This means that in this setting, even though S is a perfect surrogate for Y , δ can be substantially different from 0 and its value depends arbitrarily on the distribution of the covariate. In contrast, θ correctly captures the surrogate validity by being equal to 0.

Both settings above illustrate that the choice of discrepancy parameter is crucial, as different choices can lead to very different conclusions about surrogate validity. The rank-

based approach focuses on δ , which may not capture the causal relationship between Y and S even in simple settings. We therefore advocate focusing on θ instead.

3. Proposed Bayesian Imputation-Based Approach

BSET treats causal inference as a missing data problem, estimating parameters by recovering unobserved potential outcomes through Bayesian imputation. We leverage observed data to generate the posterior distribution of missing outcomes, creating complete datasets that enable flexible unit-level comparisons using any discrepancy metric. In accordance with the discussion in Section 2.4, we operationalize this approach by focusing on

$$V_Y = \mathbb{P}(Y_{1i} > Y_{0i}), \quad V_S = \mathbb{P}(S_{1i} > S_{0i}), \quad \theta = V_Y - V_S. \quad (5)$$

At each iteration, we impute the unobserved potential outcomes (e.g., Y_{0i} and S_{0i} for $Z_i = 0$) from their posterior predictive distribution, as detailed in Algorithm 1.

Algorithm 1 Bayesian Imputation-Based Approach.

Require: Observed data $\{Y_i, S_i, Z_i, X_i\}_{i=1}^n$, prior parameters, number of iterations T

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1: for  $t = 1$  to  $T$  do
2:   for  $i = 1$  to  $n$  do
3:     if  $Z_i = 1$  then
4:       Sample  $P_{0i}^{(t)} \sim [P_{0i} \mid D_{1:n}]$ 
5:     else
6:       Sample  $P_{1i}^{(t)} \sim [P_{1i} \mid D_{1:n}]$ 
7:     end if
8:   end for
9: end for
```

Using both the observed and imputed outcomes, we compute the probabilities in Equation (5) via Monte Carlo estimators, and obtain a posterior sample for θ by taking their

difference. Then, to evaluate S as a surrogate, we use the following hypothesis test on θ :

$$\begin{cases} H_0 : \theta \geq \eta, \\ H_1 : \theta < \eta. \end{cases} \quad (6)$$

where η is a pre-specified threshold for surrogate validation. Following Parast et al. (2024), we compute a one-sided credible interval for θ using a quantile of the T posterior draws, after discarding the first b iterations as burn-in:

$$\hat{\theta}_{1-\alpha} = \text{quantile}\left(\hat{\theta}^{(b+1:T)}, 1 - \alpha\right),$$

and we reject the null hypothesis if $\hat{\theta}_{1-\alpha} < \eta$; otherwise, we fail to reject it. Equivalently, this decision rule can be expressed in terms of the posterior probability of surrogate validity: we reject H_0 if and only if $\mathbb{P}(\theta < \eta \mid D_{1:n}) \geq 1 - \alpha$, where

$$\mathbb{P}(\theta < \eta \mid D_{1:n}) \approx \frac{1}{T-b} \sum_{t=b+1}^T \mathbb{1}(\hat{\theta}^{(t)} < \eta).$$

While this is not a Bayesian hypothesis test in the strict sense, since it does not rely on Bayes factors, it provides a natural and interpretable decision rule grounded in the posterior distribution of θ . The Bayesian decision rule is summarized in Algorithm 2.

It is clear that the primary challenge of this methodology lies in sampling the unobserved potential outcomes. For each unit i , the sampling strategy to draw the missing potential outcomes from their posterior distribution crucially depends on the assumed DGP, denoted as F_i in Equation (1). Therefore, in the next section, we propose two flexible parametric models that can be used for this sampling step.

Algorithm 2 Bayesian Decision Rule.

Require: Samples $\left\{P_{1:n}^{(t)}\right\}_{t=1}^T$, burn-in b , surrogate threshold η , significance level α

- 1: **for** $t = 1$ to T **do**
- 2: $\widehat{V}_Y^{(t)} \leftarrow \frac{1}{n} \sum_{i=1}^n \mathbb{1}(Y_{1i}^{(t)} > Y_{0i}^{(t)})$
- 3: $\widehat{V}_S^{(t)} \leftarrow \frac{1}{n} \sum_{i=1}^n \mathbb{1}(S_{1i}^{(t)} > S_{0i}^{(t)})$
- 4: $\widehat{\theta}^{(t)} \leftarrow \widehat{V}_Y^{(t)} - \widehat{V}_S^{(t)}$
- 5: **end for**
- 6: $\widehat{\theta}_{1-\alpha} \leftarrow \text{quantile}\left(\widehat{\theta}^{(b+1:T)}, 1 - \alpha\right)$
- 7: **if** $\widehat{\theta}_{1-\alpha} < \eta$ **then**
- 8: **Reject** H_0
- 9: **else**
- 10: **Fail to reject** H_0
- 11: **end if**

3.1 Parametric Models for the Data Generating Mechanism

The first model we consider excludes covariates, and therefore more closely aligns with the setting of Parast et al. (2024). Specifically, we define F_i from Equation (1) as follows:

$$\begin{aligned}
P_i &\stackrel{\text{i.i.d.}}{\sim} \mathcal{N}_4(\mu, \Sigma), \quad i = 1, \dots, n, \\
\mu &\sim \mathcal{N}_4(\mu_0, \Sigma_0), \\
\Sigma &= \text{diag}(\sigma_{1:4}) \Omega \text{diag}(\sigma_{1:4}), \\
\sigma_k &\stackrel{\text{i.i.d.}}{\sim} \text{Half-Normal}(0, s), \quad k = 1, 2, 3, 4, \\
\Omega &\sim \text{LKJ}(\tau),
\end{aligned} \tag{7}$$

where we place a Gaussian prior on the mean vector μ and we adopt a separation strategy (Barnard et al., 2000) for the covariance matrix Σ , decomposing it into a correlation matrix Ω and a scale vector $\sigma_{1:4}$. This yields interpretable priors: a Lewandowski-Kurowicka-Joe (LKJ) (Lewandowski et al., 2009) for the correlations and a half-normal for the scale.

However, one of the main advantages of our proposed Bayesian framework is its ability to naturally incorporate baseline covariates into the DGP. To illustrate this feature, we consider a second model that introduces covariates through a linear regression structure, specifying F_i from Equation (1) as follows:

$$\begin{aligned}
P_i &\stackrel{\text{i.i.d.}}{\sim} \mathcal{N}_4(BX_i, \Sigma), \quad i = 1, \dots, n, \\
B &= [\beta_1 \ \beta_2 \ \beta_3 \ \beta_4]^\top, \\
\beta_k &\stackrel{\text{i.i.d.}}{\sim} \mathcal{N}_d(\mu_\beta, \Sigma_\beta), \quad k = 1, 2, 3, 4, \\
\Sigma &= \text{diag}(\sigma_{1:4}) \Omega \text{diag}(\sigma_{1:4}), \\
\sigma_k &\stackrel{\text{i.i.d.}}{\sim} \text{Half-Normal}(0, s), \quad k = 1, 2, 3, 4, \\
\Omega &\sim \text{LKJ}(\tau),
\end{aligned} \tag{8}$$

with Gaussian priors on the entries of B and the same separation strategy for Σ as in the first model. For both models, posterior sampling is carried out via Hamiltonian Monte Carlo (Neal, 2011) using Stan (Stan Development Team, 2023b) and the `rstan` package (Stan Development Team, 2023a) in R.

3.2 Identifiability of θ and Interpretation of its Posterior

Though intuitive, the formulations in Equations (7) and (8) share a common limitation regarding identifiability. Since only one potential outcome is observed per unit, their joint distribution cannot be uniquely determined from observed data. To see how this issue manifests here, we examine the correlation matrix Ω :

$$\Omega = \begin{bmatrix} 1 & \rho_{Y_1 S_1} & \rho_{Y_1 Y_0} & \rho_{Y_1 S_0} \\ \rho_{Y_1 S_1} & 1 & \rho_{S_1 Y_0} & \rho_{S_1 S_0} \\ \rho_{Y_1 Y_0} & \rho_{S_1 Y_0} & 1 & \rho_{Y_0 S_0} \\ \rho_{Y_1 S_0} & \rho_{S_1 S_0} & \rho_{Y_0 S_0} & 1 \end{bmatrix}. \tag{9}$$

It is easy to verify that the blue entries in the correlation matrix from Equation (9) are identifiable from the data, since they involve correlations between potential outcomes that

are observed together in the data. In contrast, the remaining entries are not identifiable from the data, since they involve correlations between potential outcomes that are never jointly observed for any unit. Moreover, as shown in the general formula for θ in Web Appendix A, which was derived without assuming conditional orthogonality and homoscedasticity, the red entries directly affect the computation of θ , while the orange entries do not.

In principle, one could address this non-identifiability by imposing more structure on the correlation matrix Ω , such as setting non-identifiable entries to zero, which corresponds to assuming independent potential outcomes in this Gaussian setting. However, this is often an unreasonable restriction in practice. Therefore, it is necessary to determine how the posterior distribution of θ can be meaningfully interpreted without relying on such strong assumptions.

Under standard regularity conditions (positive prior mass on every Kullback-Leibler (KL) neighborhood of the true parameters and identifiability), Schwartz’s theorem (Schwartz, 1965) guarantees that the posterior concentrates around the true parameter value. Formal statements of this result and of the regularity conditions can be found in Web Appendix D. While our model satisfies the first condition, it lacks identifiability, meaning posterior concentration is not guaranteed. Instead, the posterior behaves as described in Gustafson (2010):

THEOREM 1 (Gustafson’s theorem (Gustafson, 2010)): *Let the true DGP be defined as*

$$D_i \stackrel{i.i.d.}{\sim} f_{\phi^*}, \quad i = 1, \dots, n,$$

where D_i is the observed data for unit i and f_{ϕ^} is the true distribution parametrized by ϕ^* . Let ϕ decompose as $\phi = (\phi_I, \phi_N)$, where ϕ_I is identifiable and ϕ_N is non-identifiable. Let the Bayesian model be defined as*

$$D_i \mid \phi \stackrel{i.i.d.}{\sim} f_{\phi_I}, \quad i = 1, \dots, n,$$

$$\phi \sim \pi,$$

where π is a prior distribution over ϕ . Then, if the Bayesian model is regular in the sense

of Definition 1 with respect to ϕ_I , the posterior distribution of ϕ_I concentrates around ϕ_I^* :

$$\pi(\phi_I \in U_I \mid D_{1:n}) \xrightarrow{P_{\phi^*}} 1,$$

for every neighborhood U_I of ϕ_I^* , where P_{ϕ^*} is the probability measure induced by f_{ϕ^*} . Whereas, the posterior distribution of ϕ_N behaves as follows:

$$\pi(\phi_N \in U_N \mid D_{1:n}) \xrightarrow{P_{\phi^*}} \pi(\phi_N \in U_N \mid \phi_I = \phi_I^*),$$

for every neighborhood U_N of ϕ_N^* , where $\pi(\phi_N \in U_N \mid \phi_I = \phi_I^*)$ is the prior distribution of ϕ_N conditional on ϕ_I being equal to its true value ϕ_I^* .

This result demonstrates that despite the lack of full identification, the posterior distribution of θ still undergoes Bayesian learning as the sample size increases. Specifically, the posterior concentrates around a set of values compatible with both the observed data and model assumptions, enabling meaningful inference for surrogate validation. However, due to the lack of concentration for the non-identifiable component, there is a non-vanishing uncertainty in the posterior distribution of θ that persists even in the asymptotic limit. While this feature of the proposed Bayesian imputation approach may appear to constitute an inherent limitation, we argue that the methodology retains merit for at least two reasons.

First, this identifiability issue is inherent to the problem rather than a specific flaw of our Bayesian approach. Any method attempting to infer the joint distribution of potential outcomes from observed data must either impose additional identifying assumptions or directly contend with this fundamental limitation. Therefore, the non-vanishing uncertainty in the posterior distribution of θ can be seen as a reflection of the inherent uncertainty in the problem of surrogate validation. However, since part of the prior uncertainty persists even asymptotically, choosing reasonable priors that reflect our beliefs about the DGP is crucial. In this regard, given the specifications chosen in Equations (7) and (8), a sensible default for the prior on Ω could be a weakly informative LKJ distribution with $\tau = 1$, which corresponds to a uniform distribution over the space of correlation matrices.

Second, though lacking asymptotic consistency, our approach may outperform alternatives that avoid non-identifiability but fail to capture the surrogate-outcome causal relationship. As shown in Section 2.3, δ can fail even in simple settings, whereas our parameter θ correctly reflects surrogate validity. This suggests our methodology may provide more accurate assessments, particularly with small samples when asymptotic limitations are less relevant.

3.3 Selection Procedure for the Validation Threshold

We now turn to selecting the surrogate validity threshold, denoted ε by Parast et al. (2024) and η in Equation (6). Establishing a principled value for ε is not obvious; thus, Parast et al. (2024) suggest choosing it to achieve a desired power for detecting treatment effects on S . We instead propose an alternative procedure for selecting η , similar in spirit but better suited to our Bayesian framework, which relies on a Bayes factor for hypothesis testing.

The intuition for our approach is that we aim to set η such that when V_S is at most η -close to V_Y , then S is a valid replacement for Y . To this end, the value of η is defined as

$$\eta = \max\{v_Y - v_S, 0\},$$

where v_Y denotes the hypothesized treatment effect on Y and v_S is defined as the value of V_S that ensures a power of $1 - \beta$ for the following hypothesis test:

$$\begin{cases} H_0 : V_S = \frac{1}{2}, \\ H_1 : V_S > \frac{1}{2}, \end{cases} \quad (10)$$

which is implemented using a Bayes factor. As an example, one plausible way to choose v_Y is to use the posterior mean of V_Y obtained from our Bayesian imputation framework.

As for v_S , we need to compute a Bayes factor for the hypothesis test in Equation (10). To do so, we first need to specify a Bayesian model taking V_S as its unknown parameter:

$$n \hat{V}_S \mid V_S \sim \text{Binomial}(n, V_S),$$

$$V_S \sim p(V_S),$$

where the prior for V_S is a Beta distribution renormalized on $(\frac{1}{2}, 1)$:

$$p(V_S) = \frac{\text{Beta}(V_S \mid a, b)}{\int_{\frac{1}{2}}^1 \text{Beta}(v \mid a, b) dv}, \quad \forall V_S \in (\tfrac{1}{2}, 1),$$

with $a, b > 0$ suitably chosen prior parameters, and $\widehat{V}_S = \frac{1}{n} \sum_{i=1}^n \mathbb{1}(S_{1i} > S_{0i})$. Note that $\widehat{V}_S$ cannot be computed directly from the observed data, since S_{1i} and S_{0i} are never jointly observed for any unit i . However, as we shall see in a moment, this is not an issue, because the procedure to select v_s only requires the sampling distribution of $\widehat{V}_S$ under the null and alternative hypothesis, rather than an observed value for it.

The resulting Bayes factor BF_n is derived and given in closed form in Web Appendix E. Then, as usual, we would like to ensure a type I error probability of α :

$$\mathbb{P}(BF_n > BF_{n,\alpha} \mid H_0) = \alpha.$$

Since $n\widehat{V}_S$ takes values in $\{0, \dots, n\}$, BF_n has at most $n+1$ distinct values, so $BF_{n,\alpha}$ can be easily obtained as the $(1 - \alpha)$ -quantile of the distribution of BF_n under H_0 , which is given in closed form in Web Appendix E. Then, given $BF_{n,\alpha}$, we define v_S as the value such that

$$\mathbb{P}\left(BF_n > BF_{n,\alpha} \mid V_S = v_S > \frac{1}{2}\right) = 1 - \beta. \quad (11)$$

It is useful to observe that the map

$$v_S \mapsto \mathbb{P}(BF_n > BF_{n,\alpha} \mid V_S = v_S)$$

is monotone on $(\frac{1}{2}, 1)$. Therefore, there exists a unique v_S satisfying Equation (11). The probability can be evaluated using the same procedure described in Web Appendix E for the distribution of BF_n given $V_S = v_S$. Since the function is continuous and monotone, v_S can be obtained using standard numerical root-finding algorithms, such as `uniroot` in R.

4. Simulation Study

We expand our assessment of the proposed Bayesian imputation-based method through a simulation study conducted across six distinct settings. In these scenarios, we evaluated

its performance relative to the frequentist rank-based approach introduced by Parast et al. (2024), focusing specifically on key metrics such as parameter coverage and statistical power.

Specifically, in Settings 1-4, we replicated the DGPs of Parast et al. (2024) to facilitate direct comparison, implementing the model in Equation (7) without covariates. Settings 1-3 feature Gaussian potential outcomes: in Setting 1, the surrogate is useless (S independent of Y), while in Settings 2 and 3, it is a linear transformation of the outcome with either negligible or more substantial Gaussian noise, representing perfect and imperfect surrogates, respectively. Setting 4 utilizes non-Gaussian distributions and a non-linear transformation with non-Gaussian noise.

In Settings 5 and 6, we purposefully break the frequentist rank-based method to highlight the advantages of our model in Equation (8). We designed these DGPs such that, conditional on covariates, potential outcomes are not identically distributed across units, thus violating a key assumption of the Mann-Whitney test. Setting 5 uses a binary covariate with parameters maximizing the gap between $\delta \approx 0.25$ and $\theta \approx 0$. Setting 6 employs a Gaussian covariate with potential outcomes following a linear model, resulting in $\delta \approx 0.35$ and $\theta \approx 0$. In both, the surrogate replicates the primary outcome with added Gaussian noise and uncorrelated potential outcomes. Full details are provided in Web Appendix F.

We ran 500 simulations for each setting, with a sample size of $n = 50$. We quantify performance in terms of coverage of the confidence/credible intervals of the true parameter (δ for the frequentist method and θ for the Bayesian method) and power, by which we mean the proportion of simulation iterations where the method deemed the surrogate to be valid. In Setting 1, the surrogate is not valid, and thus, this is the Type 1 error rate. Full details about prior parameters, and MCMC settings are given in Web Appendix F.

Results are shown in Table 1. In Settings 1 and 2, the rank-based method of Parast et al. (2024) and our Bayesian imputation approach perform similarly. In Setting 3, the Bayesian

method exhibits improved performance with higher coverage and power. Conversely, in Setting 4, where the Bayesian model is misspecified, the nonparametric rank-based approach proves more powerful. Settings 5 and 6 highlight the primary advantage of our methodology: while the frequentist method rarely identifies the surrogate as valid, our approach correctly recognizes validity in all iterations. This is expected, as these settings were designed to illustrate the limitations of the rank-based method. Our findings indicate that by adjusting for covariates, the model accurately captures surrogacy, underscoring significant power gains while clarifying that performance can be sensitive to model misspecification.

[Table 1 about here.]

5. Real Data Application

We illustrate our proposed method using data from the Diabetes Control and Complications Trial (DCCT), a multicenter, randomized study comparing an intensive treatment protocol to usual care (DCCT, 1987; Lachin and Braffett, 2021; DCCT, 1993). The intensive therapy aimed for normoglycemia via 3 or more insulin injections or an insulin pump, while usual care consisted of 1 or 2 injections per day. Our analysis focuses on a high-risk subgroup of 47 participants from the secondary-intervention cohort (those with baseline mild-to-moderate retinopathy) who had a body mass index (BMI) above 25 and a history of smoking. We define the primary outcome as the change in hemoglobin A1c (HbA1c) from baseline to 4.5 years post-randomization, with the surrogate being the earlier 1.5-year measurement.

We begin with a visual assessment of the relationship between the primary outcome and the surrogate. Figure 2 displays histograms for both variables by treatment arm; in each case, the treatment group distribution exhibits a clear rightward shift relative to the control, suggesting a beneficial effect of the intensive treatment. To further examine this relationship, Figure 3 presents a scatter plot of the surrogate against the primary outcome,

featuring fitted locally estimated scatterplot smoothing (LOESS) curves for each arm and their respective empirical cumulative distribution functions (CDFs). The scatter plot reveals that the data points are closely aligned with the diagonal, a trend further reflected by the fitted LOESS curves. This indicates that the surrogate closely mirrors the primary outcome at the individual level. Furthermore, the empirical CDFs show that the surrogate’s distribution remains consistent with that of the primary outcome within each arm, while maintaining the distinct separation between treatment groups. These visualizations provide preliminary evidence of surrogate validity, motivating the following formal assessment.

[Figure 2 about here.]

[Figure 3 about here.]

We implemented our proposed methodology to derive the posterior distribution of θ both with and without the inclusion of age and sex as covariates. For both analyses, the thresholds ε and η were determined using $\alpha = 0.05$ and $\beta = 0.3$. The resulting posterior distributions, shown in Figure 4, lead us to conclude that the surrogate is valid in both cases. For comparison, the method of Parast et al. (2024) (which does not incorporate covariates) similarly identifies the surrogate as valid.

[Figure 4 about here.]

Full numerical results for all three analyses are summarized in Table 2. While the qualitative conclusions regarding surrogacy remain consistent across all evaluated methods, the point estimates for θ , calculated as posterior means, are notably smaller than the estimates for δ obtained from the existing approach. Additionally, the inclusion of age and sex as covariates in our model leads to a further reduction in the estimated value of θ . Although the ground-truth surrogacy status remains unknown in this application, these results highlight the utility of our approach. In particular, this application demonstrates that our method

can effectively incorporate baseline information to refine the analysis, suggesting increased estimation efficiency compared to the approach of Parast et al. (2024).

[Table 2 about here.]

6. Discussion

BSET is an imputation-based Bayesian framework for surrogate evaluation that addresses limitations of the rank-based method by Parast et al. (2024), such as its lack of causal interpretation, low statistical power and the inability to adjust for covariates. By targeting a causal estimand, BSET enables inference on the individual-level association between end-points. Simulations demonstrate substantial power gains, particularly in covariate-dependent settings where rank-based methods may fail to capture valid surrogacy. These findings are supported by the DCCT application, where BSET provided smaller point estimates for the discrepancy parameter while maintaining consistent qualitative conclusions. The proposed methodology is implemented in the **BSET** R package (Carlotti and Parast, 2026). Despite these advantages, the proposed method has several limitations discussed below.

The current implementation of BSET uses a Gaussian likelihood, but this is a specific modeling choice rather than a fundamental aspect of the method. The core contribution of this work lies in the imputation-based framework, which is likelihood-agnostic in principle. A key limitation is therefore the reliance on parametric assumptions about the DGP, which are vulnerable to model misspecification, and indeed the performance of our approach deteriorates under model misspecification, as shown in the simulations. Within this framework, the likelihood can be readily replaced by alternatives better suited to the data at hand. For example, our current approach is designed for continuous outcomes and does not directly handle discrete data, but the extension to discrete outcomes can be readily achieved by changing the likelihood in the Bayesian model (e.g., using a Poisson likelihood).

To further mitigate the risk of model misspecification, future work could extend this framework to a Bayesian nonparametric setting (Ghosal and Van der Vaart, 2017). For instance, placing a Dirichlet process prior (Ferguson, 1973), perhaps centered on a Gaussian distribution, on the potential outcomes would allow relationships to be determined directly by the data. This would allow us to capture complex non-linearities or multi-modal distributions that pre-specified functional forms might overlook. However, such flexibility typically demands greater computational resources and larger sample sizes (Miller and Harrison, 2018).

A further limitation, shared with most single-trial surrogate evaluation methods, concerns transportability to future studies. The conclusion that S is a valid surrogate for Y is inherently tied to the trial’s specific population and DGP, and there is no guarantee that the same surrogacy relationship holds in future studies with different patient populations, covariate distributions, or treatment regimes. This concern partly motivated the development of meta-analytic approaches to surrogate evaluation (Buyse and Molenberghs, 1998; Burzykowski et al., 2005), which pool evidence across multiple trials to assess surrogacy more robustly across contexts. A natural and promising direction for future work is to extend BSET to a hierarchical Bayesian framework, enabling the joint modeling of surrogate-outcome relationships across multiple trials and providing a principled way to assess transportability while borrowing strength across studies.

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manuscript was not prepared under the auspices of the DCCT/EDIC study and does not reflect the views of the DCCT/EDIC study group, NIDDK-CR, or NIH. AI tools (Claude, Anthropic) were used to assist with manuscript preparation and R package development.

DATA AVAILABILITY

The DCCT data were obtained from the NIDDK-CR and are available upon request.

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SUPPLEMENTARY MATERIAL

Web Appendices A–F, referenced in Sections 2–4, are available with this paper at the Biometrics website on Oxford Academic. The BSET R package, tutorial, and replication code are available at <https://github.com/PietroCarlotti/BSET>.

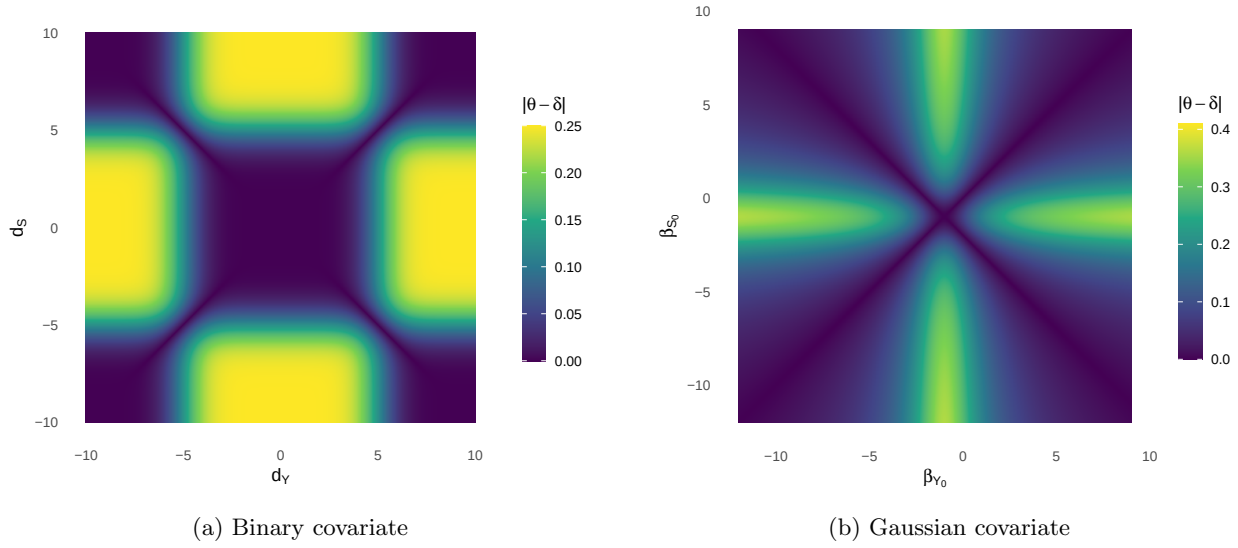


Figure 1: Left panel: plot of $|\theta - \delta|$ as a function of d_Y and d_S in the binary covariate setting, with $\Delta = 5$. Right panel: plot of $|\theta - \delta|$ as a function of β_{Y_0} and β_{S_0} in the Gaussian covariate setting, with $\Delta = 2$, $m = 2$, and $v = 2$.

Alt text: Two side-by-side heatmaps showing the absolute discrepancy between the causal estimand θ and the rank-based estimand δ as a function of two parameters, using a perceptually uniform color scale from dark purple (low discrepancy) to yellow (high discrepancy). The left panel varies d_Y and d_S in the binary covariate setting; the right panel varies the regression slopes β_{Y_0} and β_{S_0} in the Gaussian covariate setting. Both panels show that the discrepancy can be large in regions where the parameters for the primary outcome and surrogate differ substantially.

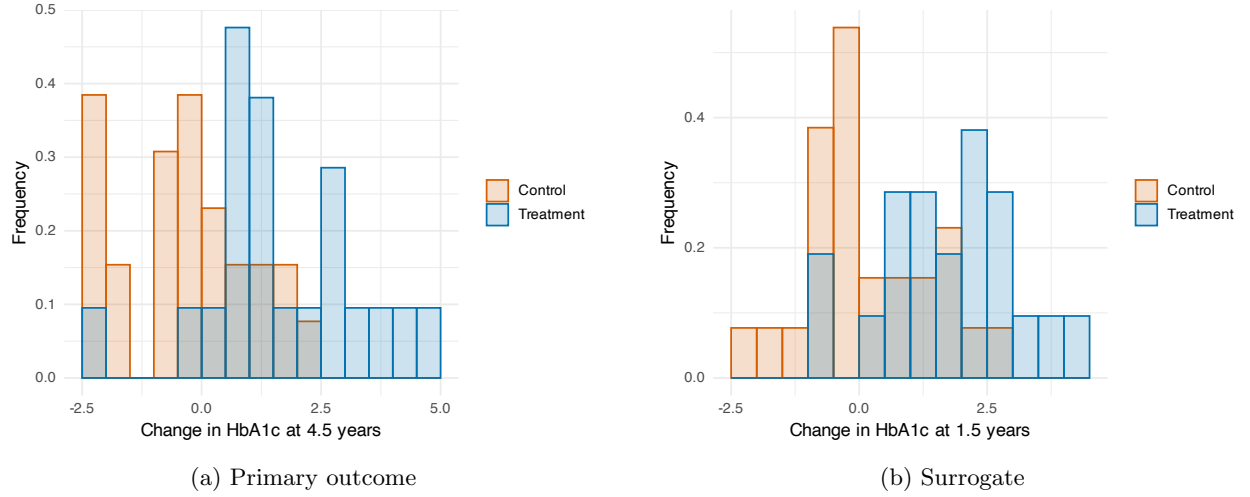


Figure 2: Left panel: histograms of the primary outcome (change in HbA1c from baseline to 4.5 years) by treatment arm. Right panel: histograms of the surrogate (change in HbA1c from baseline to 1.5 years) by treatment arm. Both panels focus on the high-risk subgroup of the DCCT dataset.

Alt text: Two side-by-side overlapping histograms comparing treatment and control groups. In both panels, the treatment group distribution (blue) is shifted to the right relative to the control group (red), indicating a beneficial treatment effect. The left panel shows change in HbA1c at 4.5 years; the right panel shows change in HbA1c at 1.5 years.

The similar shift patterns in both panels provide visual evidence that the surrogate captures the treatment effect on the primary outcome.

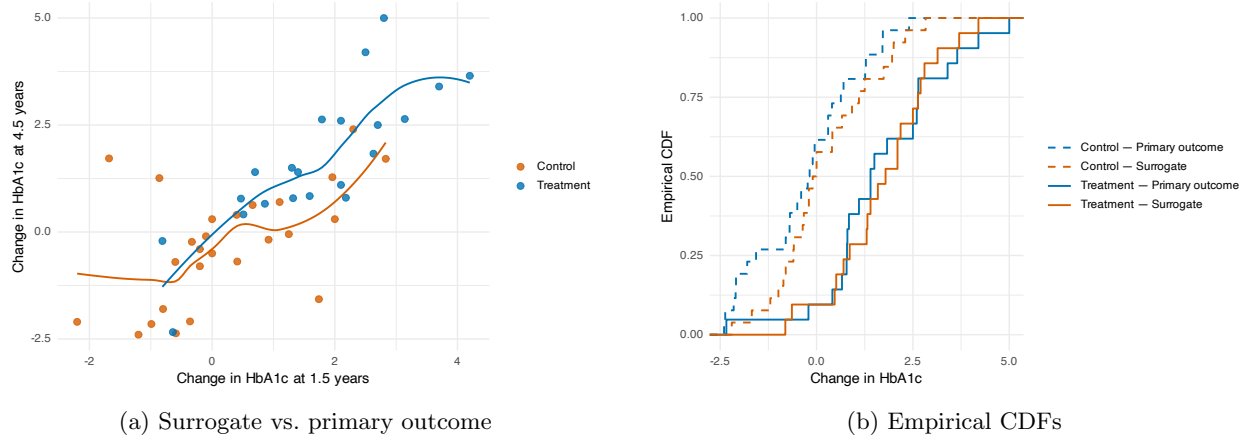


Figure 3: Left panel: scatter plot of the primary outcome (4.5-year change in HbA1c) versus the surrogate (1.5-year change in HbA1c) by treatment arm, with fitted LOESS curves. Right panel: empirical CDFs of the primary outcome and the surrogate by treatment arm.

Alt text: Left panel: scatter plot with points colored by treatment arm (blue for treatment, red for control), showing a strong positive association between surrogate and primary outcome with data points closely aligned along the diagonal; fitted LOESS curves for each arm closely follow the diagonal. Right panel: empirical CDF plot with four curves distinguished by color (blue or red) and linetype (solid or dashed) for treatment versus control and primary outcome versus surrogate; the CDF curves for the surrogate and primary outcome are closely matched within each arm.

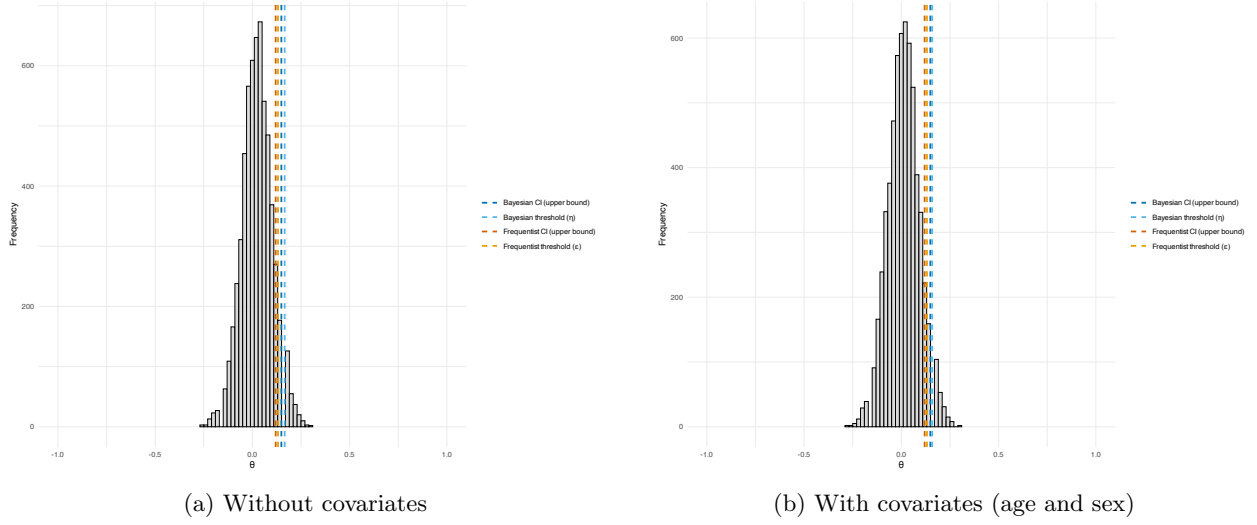


Figure 4: Left panel: posterior distribution of θ applying the method with no covariates. Right panel: posterior distribution of θ incorporating age and sex as covariates. Both panels use data from the high-risk subgroup of the DCCT.

Alt text: Two side-by-side histogram plots of the posterior distribution of theta, the discrepancy parameter. In both panels the distribution is concentrated near zero and well below the validation threshold, supporting the conclusion that the surrogate is valid. Vertical lines indicate the credible interval upper bound and the validation threshold. The right panel, incorporating age and sex as covariates, shows a slightly more concentrated posterior closer to zero compared to the left panel.

Table 1: Simulation results comparing the frequentist rank-based method and the proposed Bayesian imputation-based method in terms of coverage and power in Settings 1–6, summarized over 500 replications with $n = 50$ per replication. Bolding indicates the better performing method.

Setting	Valid Surrogate	Coverage		Power	
		Frequentist	Bayesian	Frequentist	Bayesian
1 (Useless)	No	0.98	1.00	0.00	0.00
2 (Perfect)	Yes	0.98	1.00	1.00	1.00
3 (Imperfect)	Yes	0.90	0.98	0.54	0.64
4 (Misspecified)	Yes	0.96	1.00	0.56	0.06
5 (Binary Covariate)	Yes	0.98	0.98	0.06	1.00
6 (Gaussian Covariate)	Yes	0.96	0.95	0.00	1.00

Table 2: DCCT application results. The proposed Bayesian method targets θ , while the frequentist rank-based method of Parast et al. (2024) targets δ . Thresholds are computed using $\alpha = 0.05$ and $\beta = 0.3$. A valid surrogate is concluded when the interval upper bound is below the threshold.

Method	Point Estimate	Interval Upper Bound	Threshold	Valid
BSET, no covariates	0.024	0.149	0.166	Yes
BSET, age and sex	0.012	0.149	0.158	Yes
Frequentist (Parast et al., 2024)	0.044	0.120	0.132	Yes